Evolution and development of time coding systems

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Precise temporal coding is a hallmark of both the electrosensory and auditory systems. Selective pressures to improve accuracy or encode more rapid changes have produced a suite of convergent physiological and morphological features that contribute to temporal coding. Comparative studies of temporal coding can also point to shared computational strategies, and suggest how selection might act to improve coding.

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Abbreviations

AMPA	γ-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid
EPSCs	excitatory postsynaptic currents
EPSPs	excitatory postsynaptc potentials
GABA	γ-amino butyric acid
HTC	high threshold conductance
ITD	interaural time difference
LTC	low threshold conductance
MNTB	medial nucleus of the trapezoid body
MSO	medial superior olive
NA	nucleus angularis
NL	nucleus laminaris
NM	nucleus magnocellularis
Trk	tyrosine kinase

Introduction

In the auditory system, precise encoding of temporal information has direct behavioral relevance. The timing of firing of auditory neurons carries information used for both localization and interpretation of sound. Therefore, those features of auditory neurons that lead to improved temporal processing should experience positive selection. Understanding the evolutionary and developmental events behind the similar form and function in time coding cells in birds and mammals, requires a detailed knowledge of multiple species under study, and of their phylogenetic relationships. This requires deliberate concentration on the differences between animals [1]. When we know more about the neural circuits in different groups, we can construct scenarios about how they evolved. Here, we review similar coding strategies in the time coding neurons of birds and mammals, and use these comparative studies to argue that natural selection has produced suitable convergent solutions to the problems of temporal coding.

Precise synaptic transmission

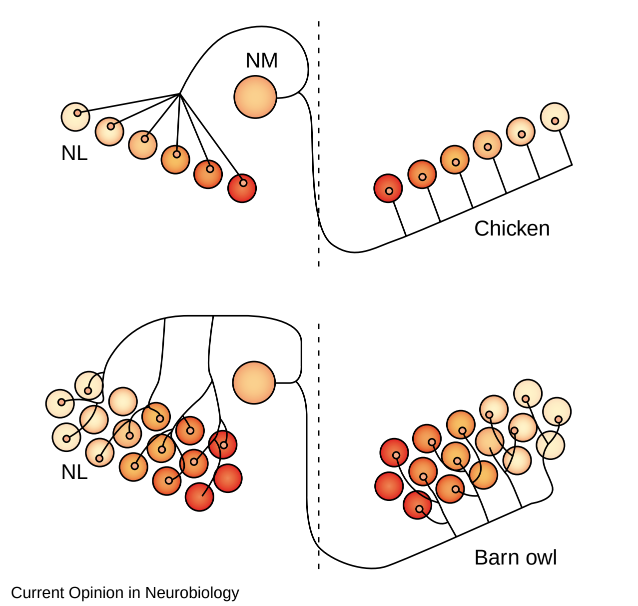
Accurate temporal coding is mediated by a variety of presynaptic and postsynaptic features [2–4]. In birds and mammals, auditory nerve afferents form endbulb synapses to ensure a secure and effective connection for the relay of the phase-locked discharges of the auditory nerve fibers to their postsynaptic targets. Endbulb synapses are also present in higher order auditory neurons of mammalian time coding pathways, including neurons of the medial nucleus of the trapezoid body (MNTB) and the type II neurons of the ventral nucleus of the lateral lemniscus [5,6].

The invasion of the presynaptic action potential into the calyx leads to the synchronous release of quanta at many sites, endowing this synapse with a high safety factor [7]. Studies of endbulbs in both the avian nucleus magnocellularis (NM) and the MNTB have identified several new features of these synapses. The invading presynaptic action potential is extremely narrow [8], probably due to rapid repolarization mediated by specific potassium conductances. Calcium influx into the presynaptic terminal is also brief and occurs only during the falling phase of the presynaptic action potential [8]. Because the action potential is narrow, its downstroke occurs quickly, as does calcium influx, reducing the synaptic delay. In addition, the brief period of calcium influx produces a confined and phasic period of neurotransmitter release, which also increases the temporal precision of transmission across the synapse [9].

Recently the rate of release at mammalian endbulb synapses has been shown to be quite rapid, of the order of 300 vesicles/sec per active zone [10•]. This rate of release is at least 10-fold higher than that recorded at hippocampal synapses [11] and is comparable to that observed at ribbon synapses [12]. At least two factors may contribute to this rapid release. First, the average size of an active zone is greater (0.12–0.18 μm^2) [13] than that at hippocampal synapses (0.04 μm^2) [14]. It has previously been postulated that larger active zones have higher release probabilities [15]. Second, a specialized mitochondrial adhesion complex is found adjacent to active zones in endbulb synapses [16]. This complex contains a vesicular chain that may feed vesicles to the active zone for rapid release similar to the 'ribbon' in retinal bipolar cells. In order to sustain the rapid release during high frequency firing induced by acoustic stimulation, the calyx also has a large pool of releasable vesicles [10•].

Transmitter release is modulated in both avian and mammalian endbulbs. The mechanisms are not the same, but all act to affect the flow of information through the endbulb. Activation of presynaptic glycine receptors enhances transmitter release in MNTB endbulbs [17•] by triggering a weakly depolarizing Cl^- current in the terminal. This depolarization enhances release by activating Ca^{2+} channels

Figure 1



Schematic diagram of time coding circuits in the chicken and barn owl. When sound comes from one side of the body, it reaches one ear before the other. The neurons of the NL translate these ITDs into sound location. The projection from the NM to the NL conforms to the model circuit proposed by Jeffress for the detection of ITDs. The model circuit is composed of two elements, delay lines and coincidence detectors. The delay lines are NM axons of varying lengths, and the coincidence detectors are NL neurons that respond maximally when they receive simultaneous inputs (i.e. when the time difference is exactly compensated for by the delay introduced by the inputs). The array of coincidence detectors receives input from the delay lines, and, because of its position in the array, each neuron responds only to sound coming from a particular direction. Thus, the anatomical place of the neuron encodes the location of the sound. These neurons compute the new variable, time difference, and transform the time code into a place code [4,66]. The colors denote the Kv3.1 gradient that follows the tonotopic distribution [31•]. Kv3.1 protein is present in NM axons innervating the NL in owl but not chicken [31•]. The selective increase of Kv3.1-like currents in the NM delay line axons in owl may contribute to the temporal synchrony necessary for accurate phase locking to high best frequencies. (Note that the tonotopic axis in reality is orthogonal to the delay lines.)

and increasing resting intraterminal Ca^{2+} concentrations. In birds, activation of presynaptic γ-amino butyric acid ($GABA_B$) receptors affects the endbulb to the NM synapse, by minimizing γ-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) receptor desensitization and therefore enhancing synaptic strength during high-frequency activity [18]. At mammalian endbulb synapses, activation of presynaptic $GABA_B$ receptors depresses singly evoked excitatory postsynaptic potentials (EPSPs) [19], but it has not yet been determined whether this depression will enhance transmission in response to rapid trains of stimulation.

‘Auditory’ AMPA receptors

Activation of AMPA receptors at endbulb synapses generates brief, large synaptic currents [3,20•]. The brevity of excitatory postsynaptic currents (EPSCs) in these neurons depends not only on the time course of release but also on the specific properties of the postsynaptic receptors. AMPA receptors in time coding auditory neurons have very rapid desensitization rates, and the duration of miniature EPSCs in auditory neurons are among the shortest recorded for any neuron [3,21]. These fortunate kinetics result from a characteristic ‘auditory’ pattern of expression [20•]. The relative abundance of the four AMPA receptor subunits (GluR1–4) has been determined by RNA analysis in chicken and rat auditory neurons. The auditory AMPA receptors in time coding neurons have very low levels of GluR2, and higher levels of the flop splice variants of GluR3 and 4 [20•,22•,23]. Quantitative expression studies of the relative abundance of GluR splice variants also show large developmental decreases in GluR2 abundance, increases in GluR3 and 4, and increases in the abundance of flop isoforms of GluRs 2–4 [20•,24]. This phenotypic switch in AMPA receptor subtype may be determined by cellular interactions during development [24,25].

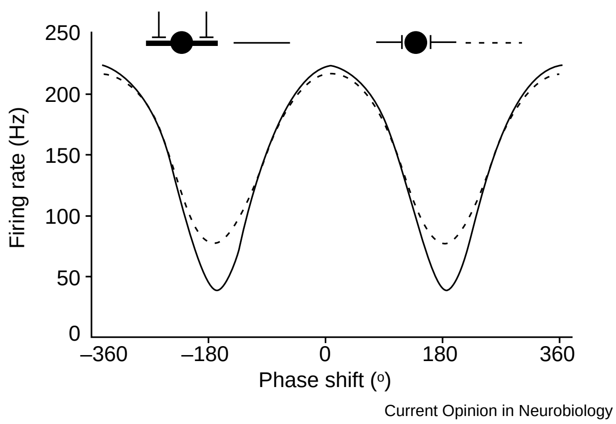
Low and high threshold potassium conductances

Intrinsic electrical properties of time coding neurons also shape their responses. At least two K^+ conductances underlie phase locked responses in auditory neurons: a low threshold conductance (LTC) and a high threshold conductance (HTC). The LTC activates at potentials near rest and is largely responsible for the outward rectification seen in many auditory neurons [2]. Activation of the LTC leads to a short active time constant so that the effects of excitation are brief and do not summate in time [26•]. Only large EPSPs reaching threshold before significant activation of the LTC would produce spikes. Blocking the LTC elicits multiple spiking in response to depolarizing current injection [3,27]. It is believed that K^+ channels underlying the LTC are composed of Kv1.1 and Kv1.2 subunits. Both subunits are expressed in auditory neurons [28•].

The HTC is characterized by an activation threshold of about –20 mV and fast kinetics [27,29]. These features of the HTC result in fast spike repolarization and a large but brief after-hyperpolarization, without influencing input resistance, threshold, or action potential rise time. Thus, the HTC can keep action potentials brief without effecting action potential generation. In addition, the HTC minimizes Na^+ channel inactivation, allowing cells to reach firing threshold sooner, and facilitates high frequency firing [29]. Currents produced by Kv3 channels share many characteristics of the HTC, and most likely underlie it. Many auditory neurons express high levels of Kv3 mRNA and protein [30,31•]. Interestingly, in several auditory nuclei, including avian NM and nucleus laminaris (NL) [31•] and rat MNTB [32], Kv3.1 protein expression varies along the tonotopic map, such that mid to high best frequency neurons are most strongly immunopositive for Kv3.1, whereas neurons with very low best frequencies are only weakly immunopositive (Figure 1). A high to low frequency gradient of Kv3.3 expression has also been observed in the electrosensory lateral line lobe of a weakly electric fish [33]. These results suggest that the electrical properties of higher order auditory neurons may vary with frequency tuning. Because no differences in either spontaneous or driven rates have been observed across the tonotopic axis, Kv3 channels may perhaps enhance the temporal precision of spike discharges.

The distribution of Kv3.1 protein in auditory neurons is largely somatic and/or axonal, consistent with its role in spike repolarization [31•,32]. Electron microscopy studies have shown that Kv3.1 is present in the membranes of endbulb terminals onto MNTB neurons, suggesting that Kv3.1 channels may be at least partially responsible for the extremely brief action potential seen at this terminal [5,6,30]. Kv3.1 protein is also present in the NM axons innervating the owl NL but not chicken NL [31•]. Kv3.1 expression in owl NM axons should reduce the width of the action potential invading the NM terminals and thus the amount of neurotransmitter released. Modeling of coincidence detectors suggests that an increase in the width of the input EPSC could impair interaural time difference (ITD) coding, which allows the localization of sound [34] (see below and Figure 1). Thus, the selective increase of Kv3.1-like currents in the NM delay line axons in the owl may contribute to the temporal synchrony necessary for accurate phase locking at high frequencies.

Figure 2



Dendritic segregation improves coincidence detection in simulated neurons. Model ITD curves for a 500 Hz stimulus showed an improvement in ITD coding when dendrites were added to the model. The contrast between the maximum spike rate (0° delay) and the minimum spike rate (180° delay) was larger when the inputs were segregated on the dendrites (solid line), compared to the point neuron model (dashed line). These ITD plots are very similar to experimental plots obtained from dog, barn owl, cat, and chicken coincidence detectors [4,67].

Encoding onsets: the appearance of similar strategies

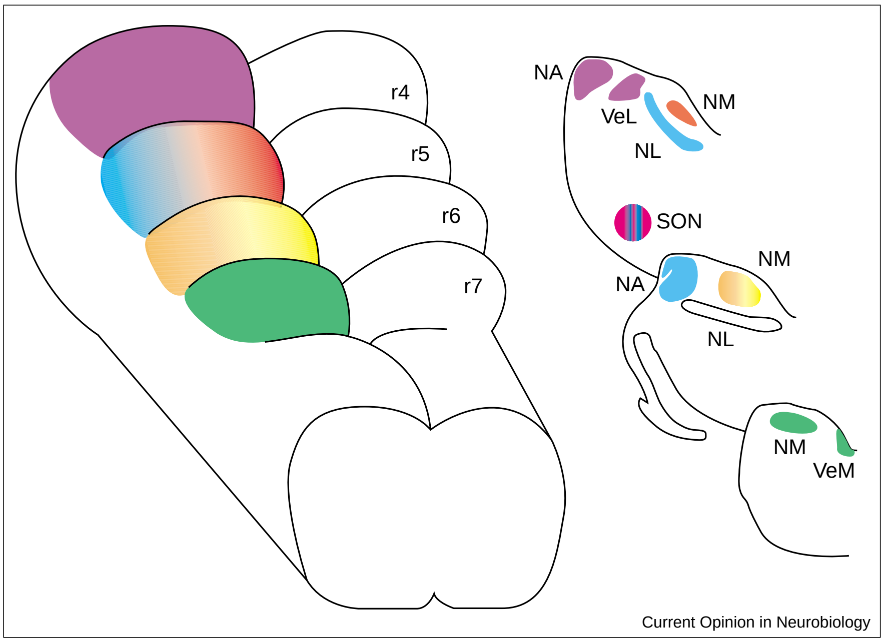
In addition to the precise temporal coding associated with phase locking, both birds and mammals have neurons that respond preferentially to onsets, or transients in sound. The computational importance of encoding onsets may be inferred by the parallel evolution of onset coding [35]. Mammals differ from birds, however, in that they have a specialized octopus cell pathway for onset coding, in addition to onset responses in other cell types [36•]. Octopus cells integrate auditory nerve inputs across a range of frequencies and encode the time structure of stimuli with great precision. Unlike the phase coding cells in the NM and the mammalian anterolateral cochlear nucleus, octopus cells transform their auditory nerve inputs to produce onset responses. How does this transformation occur? Many of the electrical and anatomical properties of octopus cells resemble those that help bushy and NM cells encode the time structure of stimuli. The brevity of the small synaptic responses in octopus cells requires the coincident activation, within 1 ms, of enough auditory nerve inputs to produce a depolarization sufficient to bring the cell to threshold [36•,37]. Like other neurons in the auditory brainstem, the intrinsic biophysical properties of octopus cells support this coincidence detection role [37]. The low input resistance of octopus cells is determined in part by a hyperpolarization-activated, mixed cation conductance and a depolarization-activated, low-threshold K^+ conductance [36•]. Also, much like the coincidence detectors in the NL and the medial superior olive (MSO), the low-threshold K^+ conductance in octopus cells allows them to be sensitive to coincident activation of their inputs, by making the membrane sensitive to fast transients in the synaptic input [37]. A rapidly rising input, such as that arising from the synchronous activation of synapses, can depolarize the membrane to threshold before the relatively slow low-threshold K^+ conductance is activated. In contrast, a slower input would fail to drive the membrane voltage to threshold, because it could not outpace this conductance [38].

Onset coding appears to have evolved in parallel in birds. In the barn owl, the chicken and the blackbird, some nucleus angularis (NA) neurons exhibit onset responses [39]. Nevertheless, there does not seem to be an avian counterpart of the octopus cell, with its frequency spanning dendritic tree [40]. Golgi analyses of both the barn owl and pigeon NA, and intracellular labeling of cells in chicken NA, have not revealed cells with thick dendrites that extend across the incoming auditory nerve inputs [40–42]. Thus, evolution of onset coding may not necessarily have produced identical solutions.

Evolution of ITD detection circuits in birds and mammals

ITD detection is a common feature of avian and mammalian auditory systems, although the resolving ability of the discrimination task varies among different species [43]. The ability of the auditory system to use ITD cues to localize sounds requires accurate encoding of temporal information. Coincidence detectors exhibit specific K^+ conductances that lead to a single or few well-timed spikes in response to a depolarizing stimulus *in vitro*. Modeling studies suggest that these fast conductances are instrumental in keeping the firing rate near zero, when the inputs are completely out of phase, but allow non-zero firing rates when the inputs are monaural [34,44].

Figure 3



The embryological origins of the cochlear nuclei were determined by injection of lipophilic dyes into small regions of the embryonic hindbrain, prior to the birth and migration of the NM. The NM arises from rhombomeres r5, r6, and r7, and the NL arises mostly from r5 with a few cells arising from r6. NM and NL thus have partially overlapping rhombomeres of origin. The cochlear NA arises from r4 [53,54•]. SON: superior olivary nucleus; VeM: medial vestibular nucleus; VeL: lateral vestibular nucleus.

Coincidence detector neurons in birds and mammals may display similar conductances and morphologies, but they are not identical. In mammals, MSO neurons do not express either Kv3.1 mRNA or protein. They do, however, express high levels of Kv3.3 message [28•,32]. Thus, differences in Kv3.1 expression between the NL and MSO structures may reflect species differences in the expression of Kv3 subfamily members. We do not know whether this variation in expression also represents a significant physiological difference, or if similar computational demands may be met by convergent expression of different gene products. A second substantial difference between birds and mammals is in the role of inhibitory inputs to coincidence detectors. In mammals, the MSO receives well-timed inhibitory input from the MNTB [45,46]. These inhibitory inputs may enhance or refine coincidence detection by increasing membrane conductance during the decay of binaural glutamatergic EPSCs. In birds, inhibitory inputs in the NL are more diffuse, and appear to decrease excitability through a gain control mechanism [46–50].

Models of coincidence detection relate form to function

A singular feature of the coincidence detectors in mammals, and low best frequency NL cells in birds, is their common morphological organization. Both are bitufted neurons with inputs from each ear segregated on the dendrites (Figure 2). Modeling studies have shown that this dendritic organization can improve coincidence detection [44]. Thus, the cell morphology and the spatial distribution of the inputs could enrich the computational power of these neurons beyond that expected from 'point neurons' or single compartment models. How does the dendritic structure of the coincidence detectors enhance their computational ability? An ITD discriminator neuron should fire when inputs from two independent neural sources coincide (or almost coincide), but not when two inputs from the same neural source coincide. A neuron that sums its inputs linearly would not be able to distinguish between these two scenarios [34]. Synaptic inputs arriving at the same dendritic compartment sum non-linearly because the driving force decreases with depolarization. Hence, the net synaptic current from several inputs arriving simultaneously at nearby sites on the same dendrite is smaller than the net current generated if these inputs are distributed on different dendrites [44,51•]. Because the bitufted structure can improve ITD detection, it is logical to suppose that ITD detection in birds and mammals evolved independently, and that their similarity is due to convergence. Evolutionary and developmental studies will be needed to determine whether or not the coincidence detector neurons in the NL and MSO shared a common bitufted ancestor, or whether they developed this morphology independently.

Advances in the development of time coding systems

Chicken embryos are widely available and accessible to experimental manipulation, and recent developmental studies have provided the first descriptions of the embryological origins of the cochlear nuclei in any vertebrate [52,53,54•,55,56]. The ITD detecting circuit in chicks is characterized by precise connections between the NM and NL, established early in development. The NM projects tonotopically to the dorsal dendrites of ipsilateral NL neurons and to the ventral dendrites of contralateral NL neurons. Tyrosine kinase (Trk) signaling may be involved in establishing these spatially segregated connections [57,58]. When NM–NL projections are forming, EphA4 receptor Trk expression in NL is asymmetric, with higher expression in the dorsal NL neuropil than in the ventral neuropil. At this time, a complementary pattern of TrkB receptor expression is observed, with higher levels of expression in the ventral neuropil. It is possible that the interplay between these two signaling systems serves to guide growing axons to the appropriate region during development, and subsequently refines the innervation pattern. Thus, developmental studies may reveal what mechanisms are responsible for the computationally appropriate segregation of inputs onto the bitufted NL neurons.

To map the embryological origins of the cochlear nuclei [52,53,54•], lipophilic dyes were injected into small regions of the embryonic hindbrain in the chicken prior to the birth and migration of the NM to show that both the NM and NL arise from a common anlage (Figure 3) [54•]. These results extended the fate mapping study of the hindbrain rhombomeres r2-r6 in quail chick chimeras [53], and show that the NM and NL have partially overlapping rhombomeres of origin. The precursors for the NM and NL are found in distinct regions within a rhombomere, however, suggesting that a lineage relationship between NM and NL cells does not exist [54•].

To aid the interpretation of the similarities and differences observed during the development of bird species, Kubke and Carr [59] have assembled a developmental time line for chickens and barn owls, constructed on the basis of common stages of structural differentiation, rather than chronological time. This type of comparison reduces the complications arising from differences in the degree of auditory specialization and from different modes of development.

Conclusions

The essential features of temporal coding are very similar in birds and mammals. Understanding the evolutionary and developmental events behind these similar features of form and function will require a detailed knowledge of multiple species, and their phylogenetic relationships. At present, this level of information is found in mormyrid electrocytes [60], frog feeding behaviour [61], eye development in cave fish [62•] and gymnotiform Na^+ channels [63•], but in few other vertebrate nervous systems.

Comparative studies of temporal coding point to shared computational principles. There are computational advantages to the bitufted NL/MSO structure. Therefore, morphological similarities do not suggest homology, but rather similar computational demands and convergence. In fact, neurons may differ in the expression and/or distribution of their ionic channels and still behave in a similar manner. Thus, there may be many acceptable ways to carry out the same computation.

Comparative studies of temporal coding also add to the discussion of whether function follows form. This issue has been previously addressed by Connors and Regehr [64]. A case can be made for the importance of form in time coding neurons of the auditory brainstem of birds and mammals, and for phase coding neurons in weakly electric fish [4].

Evolution does lead to some differences — such as the differences in Kv3.1 expression between chick and owl delay lines. How do these emerge? Nishikawa [65] has noted that the 'brains of animals are mixtures of innumerable small novelties against a background of conserved features'. The cladistic approach to phylogenetic analysis has revolutionized our understanding of brain evolution. Major changes in brain evolution occur only rarely, whereas small, often convergent, changes occur frequently. Furthermore, Nishikawa and others propose that some small changes in brain structure can lead to profound changes in function and behavior [65,66]. Identification of these changes in many species is needed to understand mechanisms of brain evolution.

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References and recommended reading

Papers of particular interest, published within the annual period of review, have been highlighted as:

• of special interest
•• of outstanding interest

1. Bullock TH: Comparative neuroscience holds promise for quiet revolutions. *Science* 1984, 661:473-478.

2. Oertel D: The role of timing in the brainstem auditory nuclei. *Annu Rev Physiol* 1999, 61:497-519.

3. Trussell LO: Synaptic mechanisms for coding timing in auditory neurons. *Annu Rev Physiol* 1999, 61:477-496.

4. Carr CE, M Friedman: Evolution of time coding systems. *Neural Comput* 1999, 11:1-20.

5. Brew HM, Forsythe ID: Two voltage-dependent K^+ conductances with complementary functions in postsynaptic integration at a central auditory synapse. *J Neurosci* 1995, 15:8011-8022.

6. Wu SH: Physiological properties of neurons in the ventral nucleus of the lateral lemniscus of the rat: intrinsic membrane properties and synaptic responses. *J Neurophysiol* 1999, 81:2872-2874.

7. Taschenberger H, von Gersdorff H: Fine-tuning an auditory synapse for speed and fidelity: developmental changes in presynaptic waveform, EPSC kinetics, and synaptic plasticity. *J Neurosci* 2000, 20:9162-9173.

8. Borst JG, Sakmann B: Effect of changes in action potential shape on calcium currents and transmitter release in a calyx-type synapse of the rat auditory brainstem. *Philos Trans R Soc Lond B Biol Sci* 1999, 354:347-355.

9. Sabatini BL, Regehr WG: Timing of synaptic transmission. *Annu Rev Physiol* 1999, 61:521-542.

10. • Sun JY, Wu LG: Fast kinetics of exocytosis revealed by simultaneous measurements of presynpatic capacitance and postsynaptic currents at a central synapse. *Neuron* 2001, 30:171-182.

By simultaneously measuring the presynaptic capacitance and post-synaptic current, Sun and Wu determine the size of the releasable pool (3300–5200 vesicles) and rate (>300 vesicles/sec) of synaptic vesicle release, as well as the rate of endocytosis at the endbulb or the calyx of Held. They also show that postsynaptic glutamate receptors are completely saturated by the release of only one third of the vesicles. This may explain why the size of the readily releasable pool of vesicles is much larger than that determined previously.

11. Stevens CF, Tsujimoto T: Estimates for the pool size of releasable quanta at a single central synapse and for the time required to refill the pool. *Proc Natl Acad Sci USA* 1995, 92:846-849.

12. Von Gersdorff H: Synaptic ribbons: versatile signal transducers. *Neuron* 2001, 29:7-10.

13. Lubke JHR, Satzler K, Sohl L, Roth A, Eils R, Frotscher M, Sakmann B: The calyx of Held: 3D reconstruction and quantitative analysis of its morphology. *Soc Neurosci Abstr* 2000, 20:676.

14. Schikorski T, Stevens CF: Quantitative ultrastructural analysis of hippocampal excitatory synapses. *J Neurosci* 1997, 17:5858-5867.

15. Walmsley B, Alvarez FJ, Fygge RE: Diversity of structure and function at mammalian central synapses. *Trends Neurosci* 1998, 21:81-88.

16. Rowland KC, Irby NK, Spirou GA: Specialized synapse-associated structures within the calyx of Held. *J Neurosci* 2000, 20:9135-9144.

17. • Turecek R, Trussell LO: Presynaptic glycine receptors enhance transmitter release at a mammalian central synapse. *Nature* 2001, 411:587-590.
The authors show that, unlike the classical action of glycine, presynaptic glycine receptors trigger a weakly depolarizing Cl^- current in the nerve terminal. The depolarization enhances transmitter release by activating Ca^{2+} channels and increasing resting intraterminal Ca^{2+} concentrations. These results indicate that under certain conditions presynaptic Cl^- channels may increase transmitter release.

18. Brenowitz S, Trussell LO: Minimizing synaptic depression by control of release probability. *J Neurosci* 2001, 21:1857-1867.

19. Isaacson JS: GABAB receptor-mediated modulation of presynaptic currents and excitatory transmission at a fast central synapse. *J Neurophysiol* 1998, 80:1571-1576.

20. • Parks TN: The AMPA receptors of auditory neurons. *Hear Res* 2000, 147:77-91.
This paper provides a review of the localization of GluR1–4 mRNA and immunoreactivity in mammals and birds, and of the involvement of AMPA receptors in synaptic transmission in the auditory system. Recent biochemical and electrophysiological studies of AMPA receptors in brainstem auditory neurons are also reviewed, along with data concerning how these properties emerge during normal development.

21. Gardner S, LO Trussell, D Oertel: Time course and permeation of synaptic AMPA receptors in cochlear nucleus neurons correlate with input. *J Neurosci* 1999, 19:8721-8729.

22. • Ravindranathan A, Donevan SD, Sugden SG, Greig A, Rao MS, Parks TN: Contrasting molecular composition and channel properties of AMPA receptors on chick auditory and brainstem motor neurons. *J Physiol* 2000, 523:667-684.
This study compares AMPA receptors in the chick ascending auditory pathway with those on regular brainstem motor neurons. Auditory AMPA receptors share a number of structural and functional properties that include a high content of GluR3 flop and GluR4 flop subunits, with a high permeability to Ca^{2+} from selective post-transcriptional suppression of GluR2 expression.

23. Schmid S, Guthman A, Ruppersberg JP, Herbert H: Expression of AMPA receptor subunit Flip/Flop splice variants in the rat auditory brainstem and inferior colliculus. *J Comput Neurol* 2001, 430:160-171.

24. Marianowski R, Liao WH, Van Den Abbeele T, Fillit P, Herman P, Frachet B, Tran Ba Huy P: Expression of NMDA, AMPA and $GABA_A$ receptor subunit mRNAs in the rat auditory brainstem. I. Influence of early auditory deprivation. *Hear Res* 2000, 150:1-11.

25. Lawrence JJ, Trussell LO: Long-term specification of AMPA receptor properties after synapse formation. *J Neurosci* 2000, 20:4864-4870.

26. • Bal R, Oertel D: Hyperpolarization-activated, mixed-cation current I(h) in octopus cells of the mammalian cochlear nucleus. *J Neurophysiol* 2000, 842:806-817.
This paper describes the anatomical and biophysical specializations of octopus cells that allow them to detect the coincident firing of groups of auditory nerve fibers and to convey the precise timing of that coincidence to their targets. Both anatomical and biophysical specializations make octopus cells detectors of the coincident firing of their auditory nerve fiber inputs.

27. Rathouz M, Trussell LO: A characterization of outward currents in neurons of the nucleus magnocellularis. *J Neurophysiol* 1998, 80:2824-2835.

28. • Grigg JJ, Brew HM, Tempel BL: Differential expression of voltage gated potassium channel genes in auditory nuclei of the mouse brainstem. *Hear Res* 2000, 140:77-90.
This mouse study shows high levels of expression of Kv1.1 and very low level expression of Kv3.1 in the octopus cell area of the cochlear nucleus and in the lateral superior olivary nucleus. The unique expression patterns suggest that Kv channel gene expression is regulated to allow brainstem auditory neurons to transmit temporally patterned signals with high fidelity.

29. Wang LY, Gan L, Forsythe ID, Kaczmarek LK: Contribution of the Kv3.1 potassium channel to high-frequency firing in mouse auditory neurones. *J Physiol (Lond)* 1998, 509:183-194.

30. Perney TM, Kaczmarek LK: Localization of a high threshold potassium channel in the rat cochlear nucleus. *J Comp Neurol* 1997, 386:178-202.

31. • Parameshwaran S, Carr CE, Perney TM: Expression of the Kv3.1 potassium channel in the avian auditory brainstem. *J Neurosci* 2001, 21:485-494.
Here, we show that the major difference in localization of Kv3.1 protein between barn owls and chickens is the expression of Kv3.1 in the NM axons and terminals in the region of the barn owl NL. This location of Kv3.1 channels supports its postulated function in reducing the width of action potentials as they invade the presynaptic terminal. The presynaptic localization may be a specialization for enabling neurons in owl NM to transmit high-frequency temporal information with little jitter.

32. Li W, Kaczmarek LK, Perney TM: The localization of two high threshold potassium channels subunits in the rat auditory brainstem. *J Comp Neurol* 2001, 437:196-218.

33. Rashid AJ, Morales E, Turner RW, Dunn RJ: The contribution of dendritic Kv3 K^+ channels to burst threshold in a sensory neuron. *J Neurosci* 2001, 21:125-135.

34. Simon JZ, Carr CE, Shamma SA: A dendritic model of coincidence detection in the avian brainstem. *Neurocomput* 1999, 26-7:263-269.

35. Carr CE, Code RA: Anatomy and physiology of the central auditory system of birds and reptiles. In: *Comparative Hearing: Birds and Reptiles.* Edited by Popper A, Fay R, Dooling R. Berlin; Springer-Verlag: 2000.

36. • Oertel D, Bal R, Gardner S, Smith P, Joris P: Detection of synchrony in the activity of auditory nerve fibers by octopus cells of the mammalian cochlear nucleus. *Proc Natl Acad Sci USA* 2000, 97:11773-11779.
The anatomical and biophysical specializations of octopus cells allow them to detect the coincident firing of groups of auditory nerve fibers and to convey the precise timing of that coincidence to their targets. Their low input resistance makes individual EPSPs brief, so that they must be generated within 1 ms in order to sum, and their voltage-sensitive conductances prevent firing if the activation of auditory nerve inputs is not sufficiently synchronous.

37. Golding NL, Ferragamo MJ, Oertel D: Role of intrinsic conductances underlying responses to transients in octopus cells of the cochlear nucleus. *J Neurosci* 1999, 19:2897-2905.

38. Kalluri, Delgutte B: Mathematical models of cochlear nucleus onset neurons. I. Point neuron with many weak synaptic inputs. *J Comput Neurosci* 2001, in press.

39. Köppl C, Carr CE, Soares D: Diversity of response patterns in the cochlear nucleus angularis (NA) of the barn owl. *ARO Abstr* 2001.

40. Soares D, Carr CE: The cytoarchitecture of the nucleus angularis of the barn owl (***Tyto alba***). *J Comp Neurol* 2001, 429:192-205.

41. Häusler UH, Sullivan WE, Soares D, Carr CE: A morphological study of the cochlear nuclei of the pigeon (***Columba livia***). *Brain Behav Evol* 1999, 54:290-302.

42. Soares D, Hyson RL, Code RA, Carr CE: The intrinsic physiological and morphological characteristics of the chicken nucleus angularis. *Soc Neurosci Abstr* 2001.

43. Heffner RS: Comparative study of sound localization and its anatomical correlates in mammals. *Acta Oto-Laryngol Suppl* 1997, 532:46-53.

44. Agmon-Snir H, Carr CE, Rinzel J: The role of dendrites in auditory coincidence detection. *Nature* 1998, 393:268-272.

45. Smith AJ, Owens S, Forsythe ID: Characterisation of inhibitory and excitatory postsynaptic currents of the rat medial superior olive. *J Physiol* 2000, 529:681-698.

46. Grothe B, Klump GM: Temporal processing in sensory systems. *Curr Opin Neurobiol* 2000, 10:467-473.

47. Monsivais P, Yang L, Rubel EW: GABAergic inhibition in nucleus magnocellularis: implications for phase locking in the avian auditory brainstem. *J Neurosci* 2000, 20:2954-2963.

48. Yang L, Monsivais P, Rubel EW: The superior olivary nucleus and its influence on nucleus laminaris: a source of inhibitory feedback for coincidence detection in the avian auditory brainstem. *J Neurosci* 1999, 19:2313-2325.

49. Funabiki K, Koyano K, Ohmori H: The role of GABAergic inputs for coincidence detection in the neurones of nucleus laminaris of the chick. *J Physiol* 1998, 508:851-869.

50. Pena JL, Viete S, Albeck Y, Konishi M: Tolerance to sound intensity of binaural coincidence detection in the nucleus laminaris of the owl. *J Neurosci* 1996, 16:7046-7054.

51. • Reyes AD: Influence of dendritic conductances on the input-output properties of neurons. *Annu Rev Neurosci* 2001, 24:653-675.
This review of voltage- and time-dependent ion channels in dendrites presents current thinking about the relationship between synaptic input and neuronal firing. Traditionally, dendrites have been viewed as passive structures whose main function is to funnel synaptic input into the soma. However, dendrites contain a wide variety of voltage- and time-dependent ion channels. These channels can alter the amplitude and time course of the synaptic input and transform the original signal. Thus, signal processing in dendrites needs careful study.

52. Cambronero F, Puelles L: Rostrocaudal nuclear relationships in the avian medulla oblongata: a fate map with quail chick chimeras. *J Comp Neurol* 2000, 427:522-545.

53. Marin F, Puelles L: Morphological fate of rhombomeres in quail/chick chimeras: a segmental analysis of hindbrain nuclei. *Eur J Neurosci* 1995, 7:1714-1738.

54. • Cramer KS, Fraser SE, Rubel EW: Embryonic origins of auditory brain-stem nuclei in the chick hindbrain. *Dev Biol* 2000, 224:138-151.
The auditory nuclei of the chick brain stem (NM and NL) arise from a common anlage, which differentiates as the cells migrate and acquire their mature morphologies. The authors injected lipophilic dyes into small regions of the embryonic hindbrain and showed that NM arises from rhombomeres r5, r6, and r7, and NL arises mostly from r5 with a few cells arising from r6. Thus, there is not a lineage relationship between NM and NL cells.

55. Hendriks R, Morest DK, Kaczmarek LK: Shaw-like potassium currents in the auditory rhombencephalon throughout embryogenesis. *J Neurosci Res* 1999, 58:791-804.

56. Hendriks R, Morest DK, Kaczmarek LK: Role in neuronal cell migration for high-threshold potassium currents in the chicken hindbrain. *J Neurosci Res* 1999, 58:805-814.

57. Cochran S, Stone JS, Bermingham-McDonogh O, Akers SR, Lefcort F, Rubel E: Ontogenetic expression of Trk neurotrophin receptors in the chick auditory system. *J Comp Neurol* 1999, 413:271-288.

58. Cramer KS, Rosenberger MH, Frost DM, Cochran SL, Pasquale EB, Rubel EW: Developmental regulation of EphA4 expression in the chick auditory brainstem. *J Comput Neurol* 2000, 426:270-278.

59. Kubke MF, Carr CE: Development of the auditory brainstem of birds: comparison between barn owls and chickens. *Hear Res* 2000, 147:1-20.

60. Sullivan JP, Lavoue S, Hopkins CD: Molecular systematics of the african electric fishes (***Mormyroidea***: ***teleostei***) and a model for the evolution of their electric organs. *J Exp Biol* 2000, 203:665-683.

61. Nishikawa KC: Neuromuscular control of prey capture in frogs. *Philos Trans R Soc Lond B* 1999, 385:941-954.

62. • Jeffery WR: Cavefish as a model system in evolutionary developmental biology. *Dev Biol* 2001, 231:1-12.
The existence of eyed surface and blind cavefish forms in *Astyanax mexicanus* allows study of the evolution of developmental mechanisms. The polarity of evolutionary changes are known, and several different cavefish populations have evolved constructive and regressive changes independently. Although adult cavefish lack functional eyes, small eye primordia are formed during embryogenesis. Furthermore, when a surface fish lens is transplanted into a cavefish optic cup, eyes are formed, indicating that cavefish optic tissues have conserved the ability to respond to lens signaling.

63. • Lopreato GF, Lu Y, Southwell A, Atkinson NS, Hillis DM, Wilcox TP, Zakon HH: Evolution and divergence of sodium channel genes in vertebrates. *Proc Natl Acad Sci USA* 2001, 98:7588-7592.
Invertebrate species possess one or two Na^+ channel genes, yet there are 10 in mammals. Here, phylogenetic comparisons show that each teleost gene is orthologous to a Na^+ channel gene or gene cluster on a different mammalian chromosome, supporting the hypothesis that four Na^+ channel genes were present in the ancestors of teleosts and tetrapods.

64. Connors BW, Regehr WG: Neuronal firing: does function follow form? *Curr Biol* 1996, 6:1560-1562.

65. Nishikawa KC: Emergence of novel functions during brain evolution. *Bioscience* 1997, 47:341-354.

66. Katz PS, Harris-Warrick RM: The evolution of neuronal circuits underlying species-specific behavior. *Curr Opin Neurobiol* 1999, 9:628-33.

67. Joris PX, Smith PH, Yin TCT: Coincidence detection in the auditory system: 50 years after Jeffress. *Neuron* 1998, 21:1235-1238.